

DATA MAPPING FOR TELECOM CHURN BY JOHN FAFIOLU

1. Defining the Problem & Scope Before looking at any data, we need to define what "churn" actually means for this telecom business.

- **What counts as churn?** We should only look at *voluntary* churn—customers who actively chose to cancel their subscription. We need to exclude customers whose accounts were closed simply because a credit card expired, as that is a separate billing issue.
- **The Goal:** We need to figure out exactly what happened last quarter (Descriptive) and look for behavioral triggers indicating who might leave in the next 30 days (Predictive).

2. The Data Requirements If I were asking for a dataset to analyze this problem, I would request a table containing the following fields for each customer:

- **Profile Data:** Customer_ID, Region, and Plan_Type (e.g., Basic or Premium).
- **Subscription Timeline:** Join_Date, Cancel_Date (if applicable), and Current_Status (Active or Canceled).
- **Behavioral Data:** Average_Monthly_Data_Used, Data_Used_Last_7_Days, and Number_of_Support_Calls_Last_30_Days.

3. Tackling the Descriptive Question *Question: What was last quarter's churn rate, by region and plan type?*

To solve this, I would follow these logical steps:

- **Isolate the Starting Cohort:** Filter the dataset to include *only* the customers who were actively subscribed on the very first day of the previous quarter.
- **Count the Churners:** Out of that specific starting group, count how many customers changed their status to "Canceled" by the last day of the quarter.
- **Calculate the Rate:** Divide the number of canceled customers by the starting number of customers to get the churn percentage.
- **Group the Results:** Sort these percentages by the Region and Plan_Type columns to identify which areas or plans are losing the most customers.

4. Tackling the Predictive Question *Question: Which currently active customers are most likely to cancel in the next 30 days?*

Static information like a customer's region won't tell us if they are about to leave. Instead, we need to look at their recent behavior for triggers. I would build a simple risk-flagging system based on two main indicators:

- **The Usage Drop-off:** If a customer typically uses 50GB of data a month, but their `Data_Used_Last_7_Days` suddenly drops to near zero, they have likely stopped using the service and are preparing to cancel.
- **The Frustration Spike:** If a customer's `Number_of_Support_Calls_Last_30_Days` jumps from zero to three, they are experiencing major friction (like recurring network outages).
- **The Output:** I would filter the list of currently active customers to flag anyone exhibiting these behavioral triggers. This "High-Risk List" can then be handed over to the retention team so they can proactively reach out or offer a discount before the customer officially hits the cancel button.